

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Social_Media_Engagement.csv to Social_Media_Engagement.csv

```
df = pd.read_csv("Social_Media_Engagement.csv")
```

```
print("First 5 rows:")
print(df.head())
```

First 5 rows:

	Account ID	Username	Platform	Follower Count	Posts Per Week	\
0	1	harrisalisa	TikTok	54217	3	
1	2	rhicks	LinkedIn	987518	5	
2	3	qthomas	Facebook	218870	3	
3	4	carlosholt	Instagram	207432	6	
4	5	parsonsashley	Facebook	350204	2	

	Engagement Rate	Ad Spend (USD)	Conversion Rate	Campaign Reach
0	0.0986	538.10	0.0490	1308
1	0.0834	479.24	0.0174	13302
2	0.1020	150.36	0.0318	11043
3	0.0834	932.62	0.0400	12074
4	0.0642	504.44	0.0463	14083

```
print("\nDataset Info:")
print(df.info())
```

```
print("\nStatistical Summary:")
print(df.describe())
```

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 10016 entries, 0 to 10015

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Account ID	10016 non-null	int64
1	Username	10016 non-null	object
2	Platform	10016 non-null	object
3	Follower Count	10016 non-null	int64
4	Posts Per Week	10016 non-null	int64
5	Engagement Rate	10016 non-null	float64
6	Ad Spend (USD)	10016 non-null	float64
7	Conversion Rate	10016 non-null	float64
8	Campaign Reach	10016 non-null	int64

dtypes: float64(3), int64(4), object(2)

memory usage: 704.4+ KB

None

Statistical Summary:

	Account ID	Follower Count	Posts Per Week	Engagement Rate \
count	10016.000000	10016.000000	10016.000000	10016.000000
mean	4992.525559	503909.814197	3.976138	0.086725
std	2891.470351	286051.131318	1.974205	0.013330
min	1.000000	1449.000000	1.000000	0.064200
25%	2488.750000	256958.500000	2.000000	0.083400
50%	4992.500000	506605.000000	4.000000	0.085600
75%	7496.250000	749949.250000	6.000000	0.098600
max	10000.000000	999982.000000	7.000000	0.102000

	Ad Spend (USD)	Conversion Rate	Campaign Reach
count	10016.000000	10016.000000	10016.000000
mean	527.903886	0.029226	25299.707468
std	280.103530	0.011441	14119.440774
min	51.950000	0.010100	1005.000000
25%	277.400000	0.019300	13176.250000
50%	517.575000	0.028400	25255.500000
75%	783.600000	0.039200	37511.000000

```
max          998.200000      0.050000      49998.000000
```

```
# DATA CLEANING
```

```
print("\nMissing Values:")  
print(df.isnull().sum())
```

```
Missing Values:
```

```
Account ID      0  
Username        0  
Platform        0  
Follower Count  0  
Posts Per Week  0  
Engagement Rate 0  
Ad Spend (USD)  0  
Conversion Rate 0  
Campaign Reach  0  
dtype: int64
```

```
# DATA PREPROCESSING
```

```
df['Engagement Rate (%)'] = df['Engagement Rate'] * 100  
df['Conversion Rate (%)'] = df['Conversion Rate'] * 100
```

```
#DATA VISUALIZATION
```

```
#GRAPH 1: POSTS PER WEEK
```

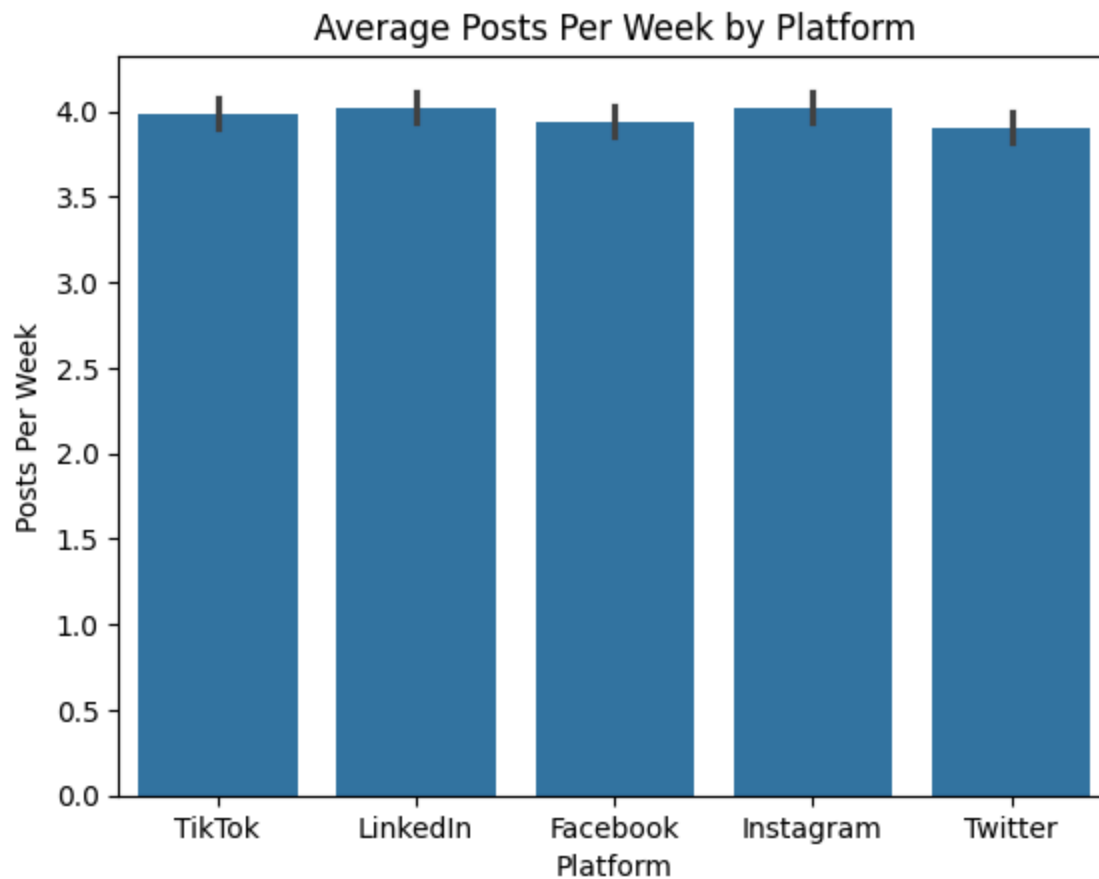
```
sns.barplot(x='Platform', y='Posts Per Week', data=df, estimator='mean')
```

```
plt.title("Average Posts Per Week by Platform")
```

```
plt.show()
```

```
print("\nObservation:")
```

```
print("Instagram and TikTok show higher average posts per week compared to other platforms, indicatir
```



Observation:

Instagram and TikTok show higher average posts per week compared to other platforms, indicating they are more active.

#GRAPH 2: ENGAGEMENT RATE

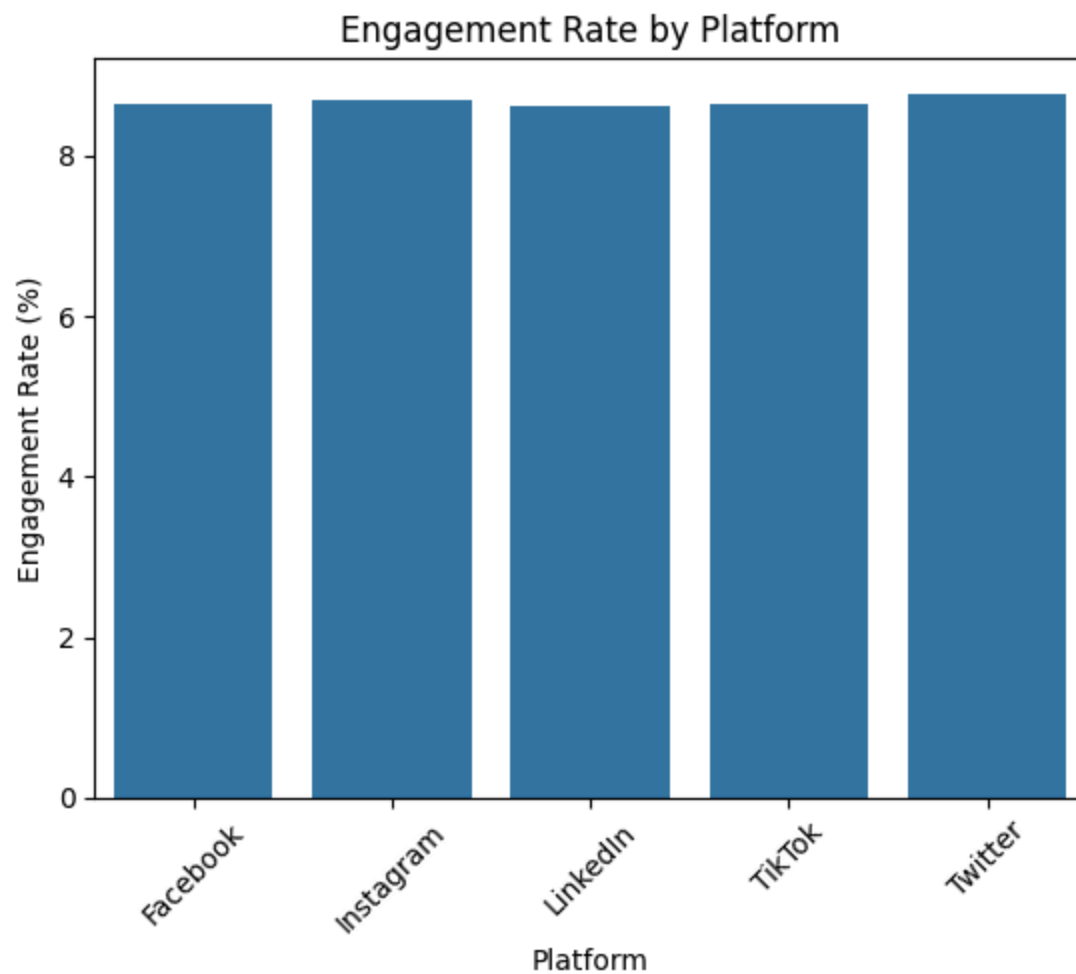
```
eng = df.groupby('Platform')['Engagement Rate (%)'].mean().reset_index()
```

```
sns.barplot(data=eng, x='Platform', y='Engagement Rate (%)')  
plt.title("Engagement Rate by Platform")  
plt.xticks(rotation=45)
```

```
plt.show()
```

```
print("\nObservation:")
```

```
print("Engagement rates are similar across platforms with slight variations.")
```



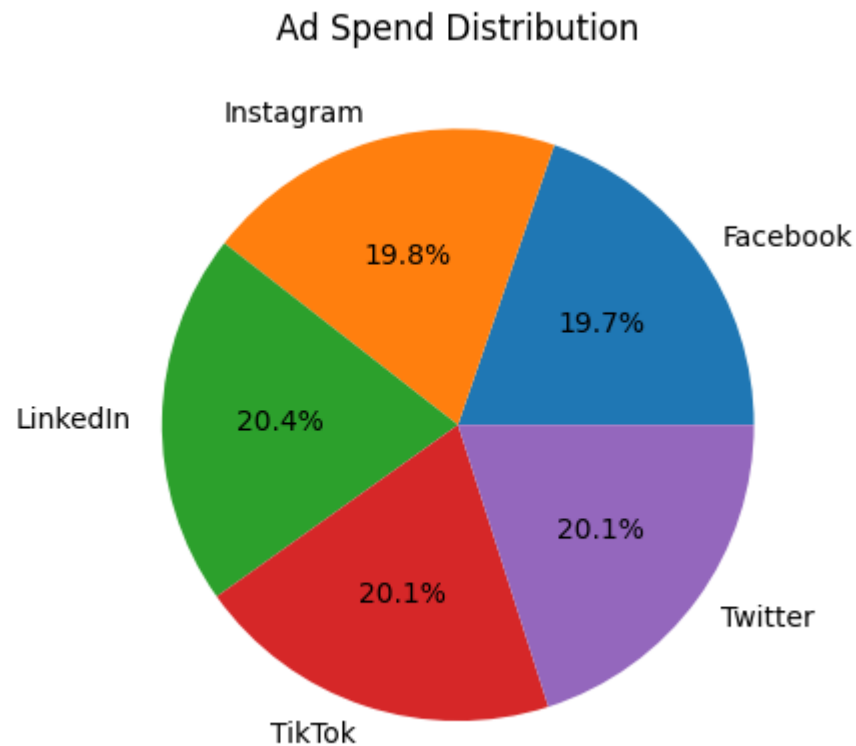
Observation:

Engagement rates are similar across platforms with slight variations.

#GRAPH 3: ADS

```
g = df.groupby('Platform')['Ad Spend (USD)'].sum()
```

```
plt.pie(g, labels=g.index, autopct='%1.1f%%')
plt.title("Ad Spend Distribution")
plt.show()
print("\nObservation:")
print("The pie chart shows that ad spend is distributed unevenly across platforms. Some platforms acc
```



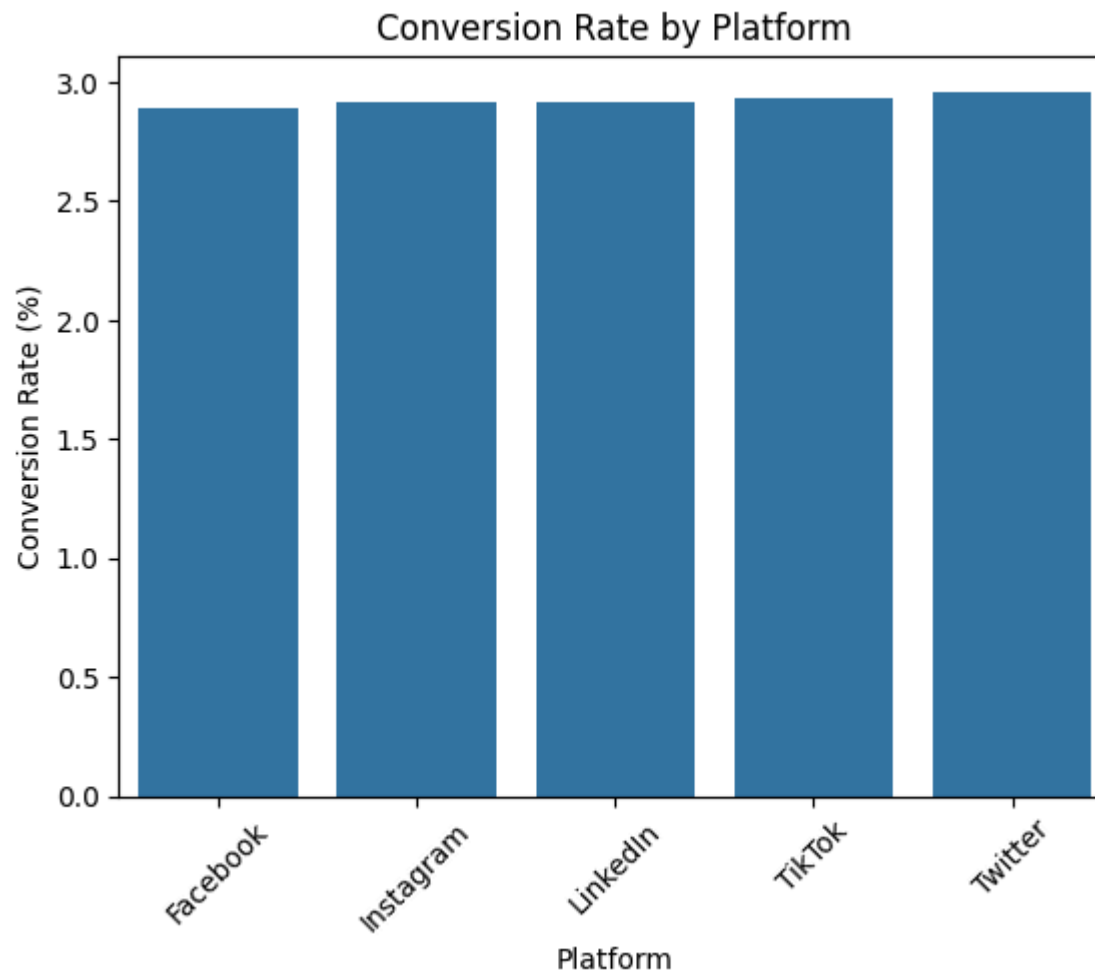
Observation:

The pie chart shows that ad spend is distributed unevenly across platforms. Some platforms account for

#GRAPH 4: CONVERSION RATE

```
conv = df.groupby('Platform')['Conversion Rate (%)'].mean().reset_index()
```

```
sns.barplot(data=conv, x='Platform', y='Conversion Rate (%)')
plt.title("Conversion Rate by Platform")
plt.xticks(rotation=45)
plt.show()
print("\nObservation:")
print("Conversion rates are relatively low, showing not all engagement leads to action")
)
```



Observation:
Conversion rates are relatively low, showing not all engagement leads to action.

```
print("\nFinal Conclusion:")
```

```
print( "FINAL CONCLUSION. ")
```

```
print("""
```

```
The analysis shows that different platforms vary in reach and activity levels.
```

```
However, engagement and conversion rates remain relatively consistent.
```

```
High reach does not guarantee high engagement or conversions.
```

```
Content quality and targeting play an important role.
```

```
""")
```

Final Conclusion:

The analysis shows that different platforms vary in reach and activity levels.

However, engagement and conversion rates remain relatively consistent.

High reach does not guarantee high engagement or conversions.

Content quality and targeting play an important role.

Start coding or [generate](#) with AI.